Rhombus and square

A parallelogram with equal sides, and the shape that is a rectangle and a rhombus at the same time.

LESSON 8–9 | 2 × 40 MIN | GEOMETRY 8, ТЕМЫ 5–6 | STAGE 9 · BEYOND

LESSON CLOCK

5' | 13' | 8' | 12' | 2'

Warm-up	5 min
Explanation	13 min
Interactive	8 min
Practice	12 min
Homework	2 min

LEARNING OBJECTIVES

- Define a rhombus and prove that its diagonals are perpendicular and bisect its angles.
- Define a square and list the properties it inherits from both parents.
- Use the area formula $S = 1/2 d_1 d_2$ for a rhombus.
- Place every quadrilateral of Chapter I correctly in the family tree.

TEXTBOOK

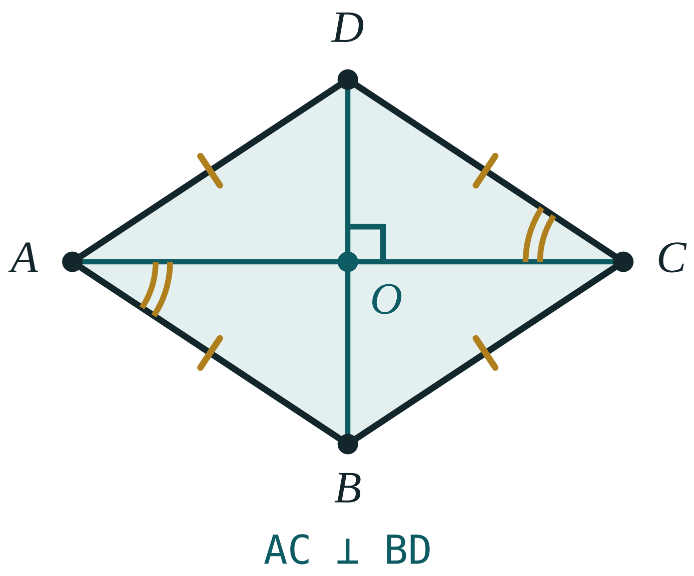
National	Темы 5–6, pp. 16–18
Cambridge	Extension beyond Stage 9
Workbook	–

01 Explanation

The rhombus

DEFINITION

A **rhombus** is a parallelogram all of whose sides are equal.



All four sides equal. The diagonals meet at right angles and each one bisects the two angles it passes through.

It inherits everything a parallelogram has, and gains two properties of its own.

THEOREM

The diagonals of a rhombus are **perpendicular**, and each diagonal **bisects** the two angles through which it passes.

Proof. $\triangle ABC$ has $AB = BC$, so it is isosceles. The diagonal BD passes through O, the midpoint of AC (parallelogram property), so BO is the median of this isosceles triangle from its apex — and therefore also its height and its angle bisector. That gives $BD \perp AC$ and $\angle ABD = \angle DBC$ at once. ∎

The area of a rhombus

The perpendicular diagonals cut the rhombus into four congruent right triangles with legs $\frac{d_1}{2}$ and $\frac{d_2}{2}$. Adding their areas:

$$S = 4 \cdot \frac{1}{2} \cdot \frac{d_1}{2} \cdot \frac{d_2}{2} = \frac{1}{2} d_1 d_2$$

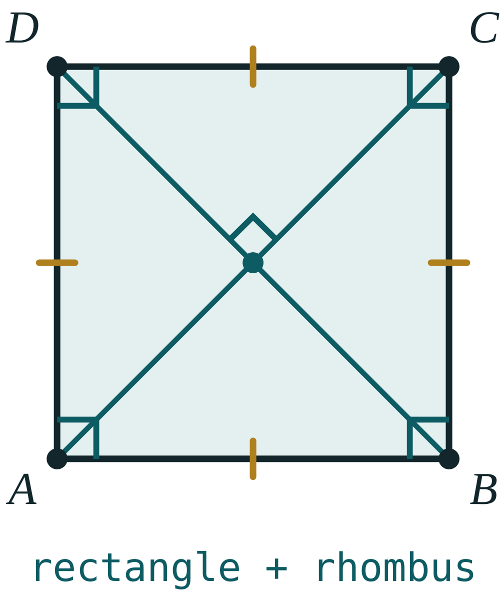
And Pythagoras in one of those triangles relates the side to the diagonals:

$$a^2 = (\frac{d_1}{2})^2 + (\frac{d_2}{2})^2$$

The square

DEFINITION

A **square** is a rectangle all of whose sides are equal — equivalently, a rhombus with a right angle.



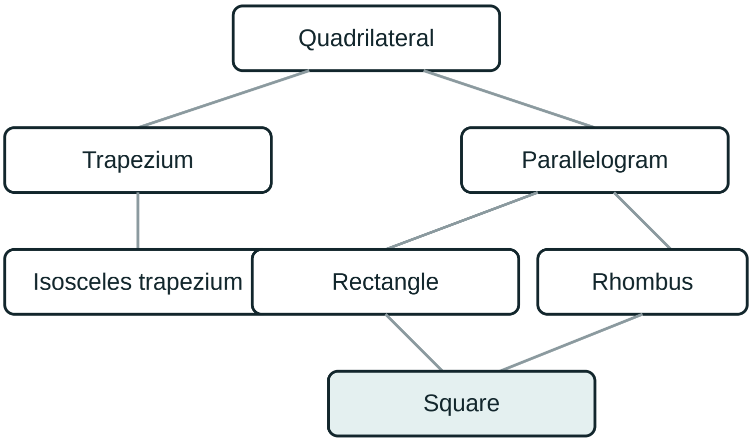
A square has every property of a rectangle and every property of a rhombus at once.

So a square has **all** of the following:

- four equal sides and four right angles;
- diagonals that bisect each other (parallelogram);
- diagonals that are **equal** (rectangle);
- diagonals that are **perpendicular** and bisect the angles into $45°$ halves (rhombus).

$$d = a\sqrt{2} \quad \cdot \quad S = a^2 = \frac{\mathrm{d}^2}{2}$$

The family tree



Each arrow means "is a special case of". The square sits at the bottom because it inherits from both the rectangle and the rhombus.

Read it downwards and every statement is true: a square *is* a rectangle, a rectangle *is* a parallelogram, a parallelogram *is* a quadrilateral. Read it upwards and almost every statement is false — that asymmetry is the point of the diagram.

02 Worked examples

EXAMPLE 1 **The diagonals of a rhombus are 12 cm and 16 cm. Find its side, perimeter and area.**

1. *half-diagonals: 6 cm and 8 cm*

   The diagonals bisect each other.

2. $a = \sqrt{6^2 + 8^2} = \sqrt{100} = 10\text{ cm}$

   Pythagoras in one of the four right triangles.

3. $P = 4 \cdot 10 = 40\text{ cm}$

   All four sides are equal.

4. $S = \dfrac{1}{2} \cdot 12 \cdot 16 = 96\text{ cm}^2$

   Half the product of the diagonals.

ANSWER

$a = 10$ cm, $P = 40$ cm, $S = 96$ cm^2

EXAMPLE 2 ***In rhombus ABCD, $\angle A = 60°$. Find $\angle ABD$ and prove that the shorter diagonal equals the side.***

1. $\angle B = 180° - 60° = 120°$

   Neighbouring angles of a parallelogram.

2. $\angle ABD = \dfrac{120°}{2} = 60°$

   The diagonal bisects the angle.

3. $\triangle ABD$ has $\angle A = 60°$ and $\angle ABD = 60°$

   so the third angle is $60°$ too.

4. $\triangle ABD$ is equilateral, so $BD = AB$

   The shorter diagonal equals the side.

ANSWER

$\angle ABD = 60°$; $BD = AB$

EXAMPLE 3 **A square has diagonal 8 cm. Find its side and area.**

1. $d = a\sqrt{2}$

   The diagonal of a square.

2. $a = \dfrac{8}{\sqrt{2}} = 4\sqrt{2}\ cm$

   Rationalise the denominator.

3. $S = \dfrac{d^2}{2} = \dfrac{64}{2} = 32\ cm^2$

   Quicker than squaring the side.

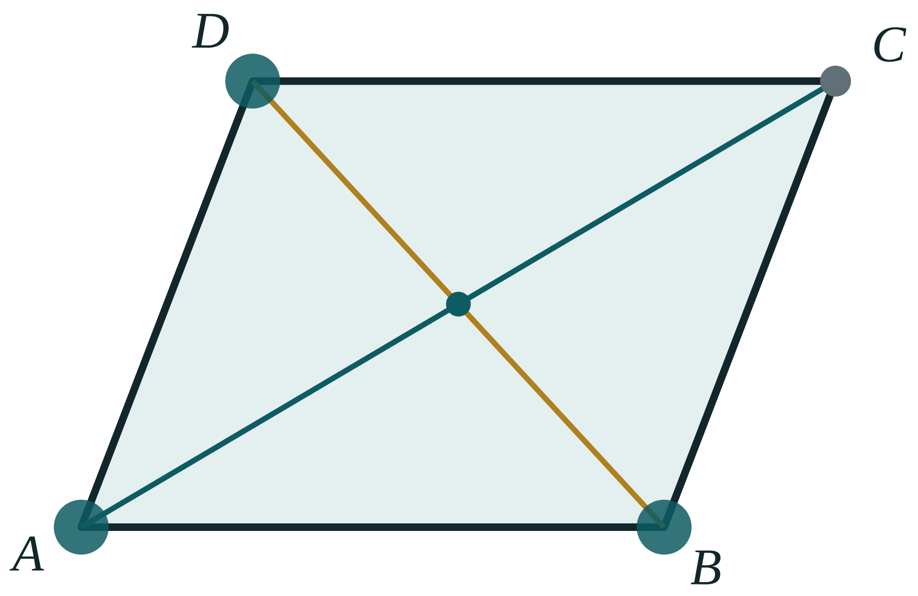
ANSWER

$a = 4\sqrt{2}\ cm \approx 5.66\ cm$, $S = 32\ cm^2$

03 Interactive model

Switch the model to rhombus mode: all four sides stay equal however you drag, and AC ⊥ BD holds throughout.

Drag the rhombus



AB **170** BC **139.3** DC **170** AD **139.3** AC **255.5** BD **176.9** AO **127.8** OC **127.8**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Rhombus	Romb	Ромб
Square	Kvadrat	Квадрат
Perpendicular	Perpendikulyar	Перпендикулярный
Bisects the angle	Burchakni teng ikkiga bo‘ladi	Делит угол пополам
Half-diagonal	Diagonalning yarmi	Половина диагонали
Equal sides	Teng tomonlar	Равные стороны
Classification	Tasnif	Классификация

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

The diagonals of a rhombus are:

- equal
- perpendicular
- equal and perpendicular
- neither

A rhombus with diagonals 6 and 8 has area:

- 48
- 24
- 14
- 10

A square is:

- a rectangle only
- a rhombus only
- both a rectangle and a rhombus
- neither

The diagonal of a square of side 5 is:

- 10
- $5\sqrt{2}$
- $\sqrt{5}$
- 25

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. *A rhombus has side 7 cm. Find its perimeter.*

   ANSWER *28 cm*

2. *The diagonals of a rhombus are 6 cm and 8 cm. Find its area.*

   ANSWER *24 cm²*

3. *In a rhombus AC = 10 cm. Find AO.*

   ANSWER *5 cm*

4. *One angle of a rhombus is 70°. Find the neighbouring angle.*

   ANSWER *110°*

5. *A square has side 6 cm. Find its perimeter and area.*

   ANSWER *24 cm, 36 cm²*

6. *A square has side 5 cm. Find its diagonal.*

   ANSWER $5\sqrt{2}$ *cm*

7. *What is the angle between the diagonals of a rhombus?*

   ANSWER *90°*

Medium · the standard question

7 PROBLEMS

1. *The diagonals of a rhombus are 12 cm and 16 cm. Find its side and perimeter.*

   ANSWER *10 cm, 40 cm*

2. *A rhombus has side 13 cm and one diagonal 10 cm. Find the other diagonal.*

   ANSWER *24 cm*

3. *In rhombus ABCD, ∠A = 60°. Find ∠ABD.*

   ANSWER *60°*

4. *A square has diagonal 8 cm. Find its side and area.*

   ANSWER $4\sqrt{2}$ cm, 32 cm²

5. *A rhombus has perimeter 40 cm and one diagonal 12 cm. Find the other.*

   ANSWER *16 cm*

6. *The area of a rhombus is 60 cm² and one diagonal is 10 cm. Find the other.*

   ANSWER *12 cm*

7. *A square has area 49 cm². Find its diagonal.*

   ANSWER $7\sqrt{2}$ cm

Hard · stretch and reason

7 PROBLEMS

1. *In rhombus ABCD, $\angle A = 60°$. Prove that the shorter diagonal equals the side.*

ANSWER $\triangle ABD$ has three $60°$ angles, so it is equilateral and $BD = AB$.

2. *A rhombus has side 10 cm and height 8 cm. Find its area, then its diagonals if one is 16 cm.*

ANSWER $S = 10 \cdot 8 = 80$; then $\frac{1}{2} \cdot 16 \cdot d_2 = 80$ gives $d_2 = 10$ cm.

3. *Prove that a parallelogram whose diagonals are perpendicular is a rhombus.*

ANSWER The diagonals bisect each other, so each half-diagonal pair gives congruent right triangles by SAS; all four sides come out equal.

4. *The diagonals of a square are 14 cm. Find its side, perimeter and area.*

ANSWER $a = 7\sqrt{2}$, $P = 28\sqrt{2}$ cm, $S = 98$ cm²

5. *A rhombus has one angle of 120° and side 6 cm. Find both diagonals.*

ANSWER Shorter = 6 cm (equilateral half), longer = $6\sqrt{3}$ cm

6. *Prove that the midpoints of the sides of a rectangle form a rhombus.*

ANSWER Each side of the new quadrilateral is a midline equal to half a diagonal, and the diagonals of a rectangle are equal — so all four sides are equal.

7. *A square and a rhombus both have side 8 cm. Compare their perimeters, and explain why their areas can differ.*

ANSWER Both perimeters are 32 cm. The rhombus can be flattened, which reduces its height and so its area; the square is the tallest case.

07 Homework

Homework — 6 problems

Geometry 8, Темы 5–6, pp. 16–18.

1. *The diagonals of a rhombus are 10 cm and 24 cm. Find its side, perimeter and area.*
2. *A rhombus has side 17 cm and one diagonal 16 cm. Find the other diagonal.*

3. *In rhombus ABCD, ∠B = 120°. Find ∠BAC.*
4. *A square has perimeter 32 cm. Find its diagonal and area.*
5. *A square has diagonal 10 cm. Find its area.*
6. *Draw the quadrilateral family tree from memory and mark where the rhombus and the square sit.*